SA1_Ramilo

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Data Mining

```r
customer_churn<- read_csv("customer_churn.csv")
```

```
## Rows: 10000 Columns: 12
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (8): CustomerID, Gender, Partner, Dependents, PhoneService, InternetServ...
## dbl (4): SeniorCitizen, Tenure, MonthlyCharges, TotalCharges
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

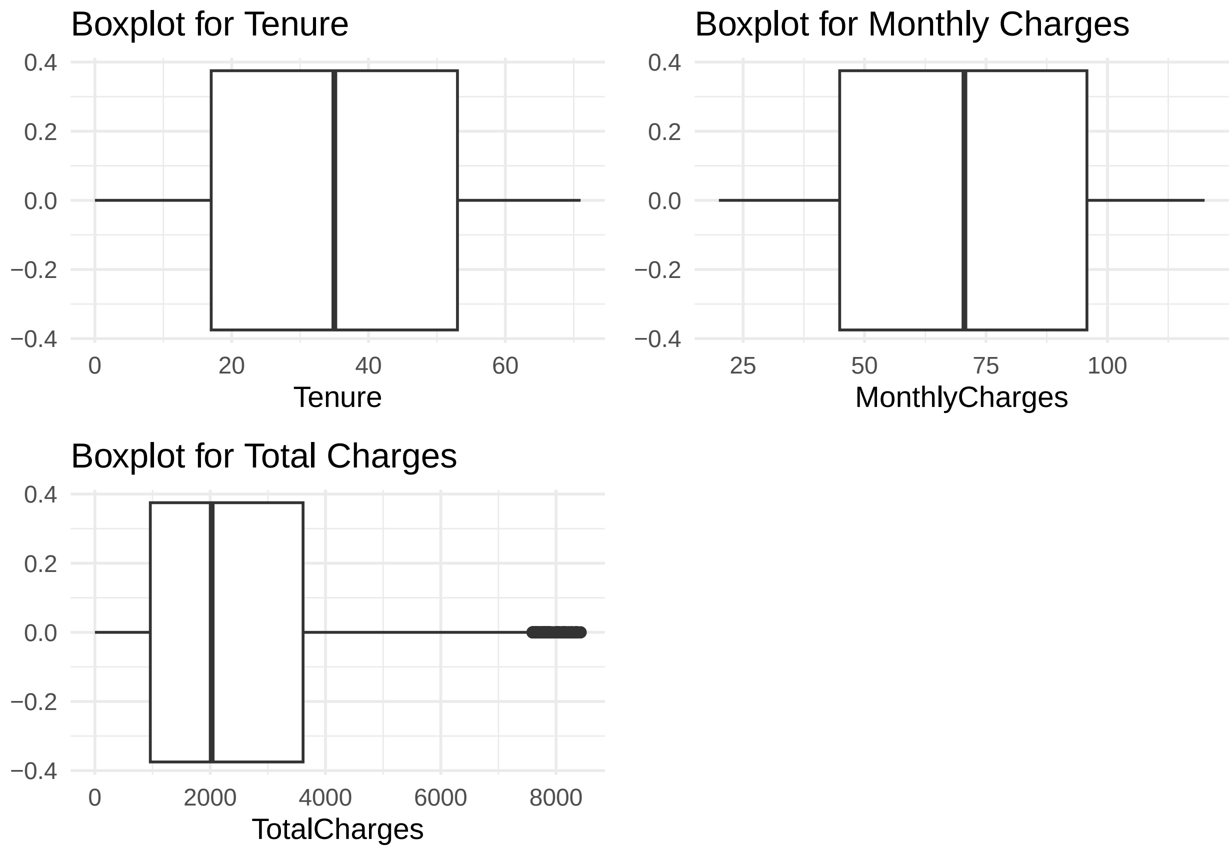
```r
summary(customer_churn)
```

```
##   CustomerID           Gender          SeniorCitizen      Partner
##  Length:10000       Length:10000       Min.   :0.0000   Length:10000
##  Class :character   Class :character   1st Qu.:0.0000   Class :character
##  Mode  :character   Mode  :character   Median :0.0000   Mode  :character
##                                        Mean   :0.1502
##                                        3rd Qu.:0.0000
##                                        Max.   :1.0000
##   Dependents            Tenure      PhoneService       InternetService
##  Length:10000       Min.   : 0.00   Length:10000       Length:10000
##  Class :character   1st Qu.:17.00   Class :character   Class :character
##  Mode  :character   Median :35.00   Mode  :character   Mode  :character
##                     Mean   :35.22
##                     3rd Qu.:53.00
##                     Max.   :71.00
##    Contract         MonthlyCharges    TotalCharges       Churn
##  Length:10000       Min.   : 20.02   Min.   :   0.0   Length:10000
##  Class :character   1st Qu.: 44.88   1st Qu.: 961.2   Class :character
##  Mode  :character   Median : 70.56   Median :2025.6   Mode  :character
##                     Mean   : 70.18   Mean   :2455.8
##                     3rd Qu.: 95.77   3rd Qu.:3611.0
##                     Max.   :119.99   Max.   :8425.6
```

Remark:

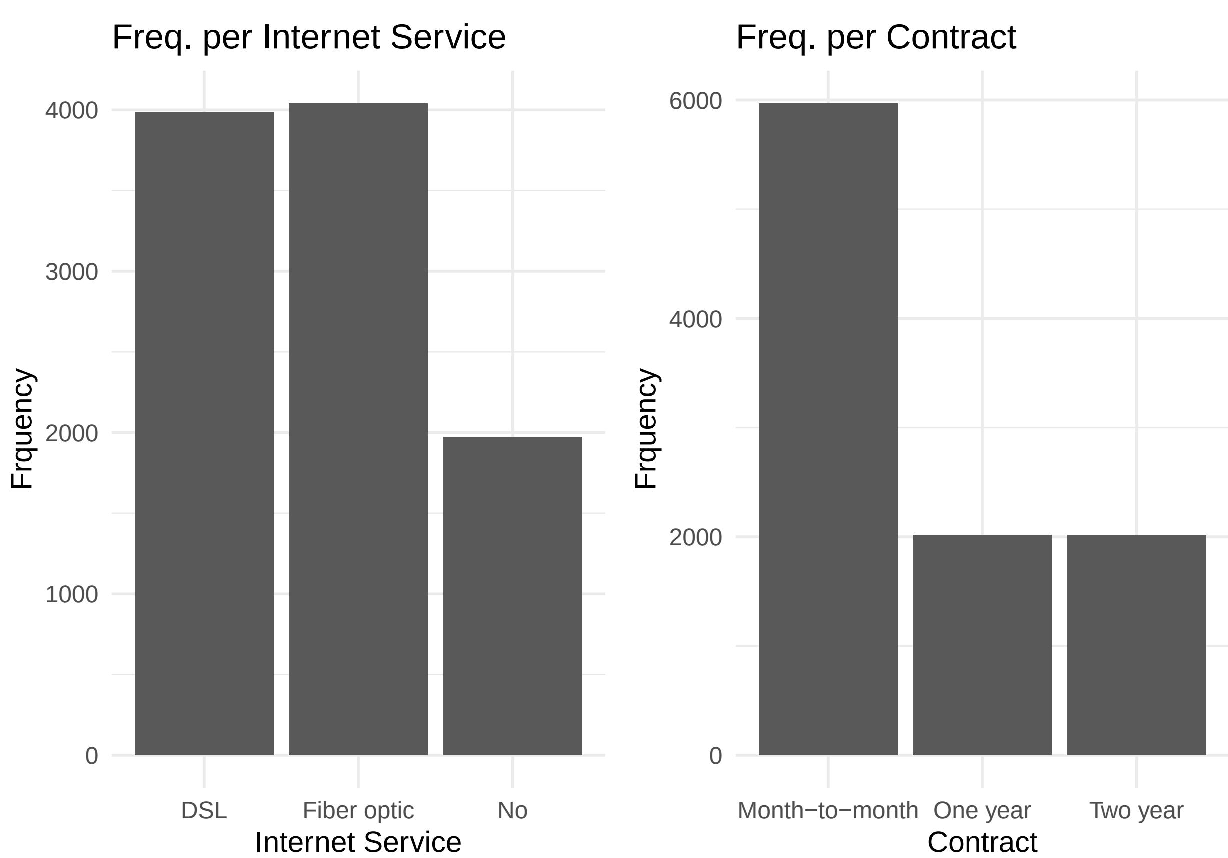
The application of data mining within this data would provide a method to identify customer churn, which is important for the business to understand its current demographic and answer who its market is.

```r
tenureBox <- ggplot(customer_churn,aes(x=Tenure))+
  geom_boxplot()+
  theme_minimal()+
  labs(
    title="Boxplot for Tenure"
  )
montlyBox <- ggplot(customer_churn,aes(x=MonthlyCharges))+
  geom_boxplot()+
  theme_minimal()+
  labs(
    title="Boxplot for Monthly Charges"
  )
totalBox <- ggplot(customer_churn,aes(x=TotalCharges))+
  geom_boxplot()+
  theme_minimal()+
  labs(
    title="Boxplot for Total Charges"
  )
boxplts <- grid.arrange(tenureBox,montlyBox,totalBox,ncol=2)
```



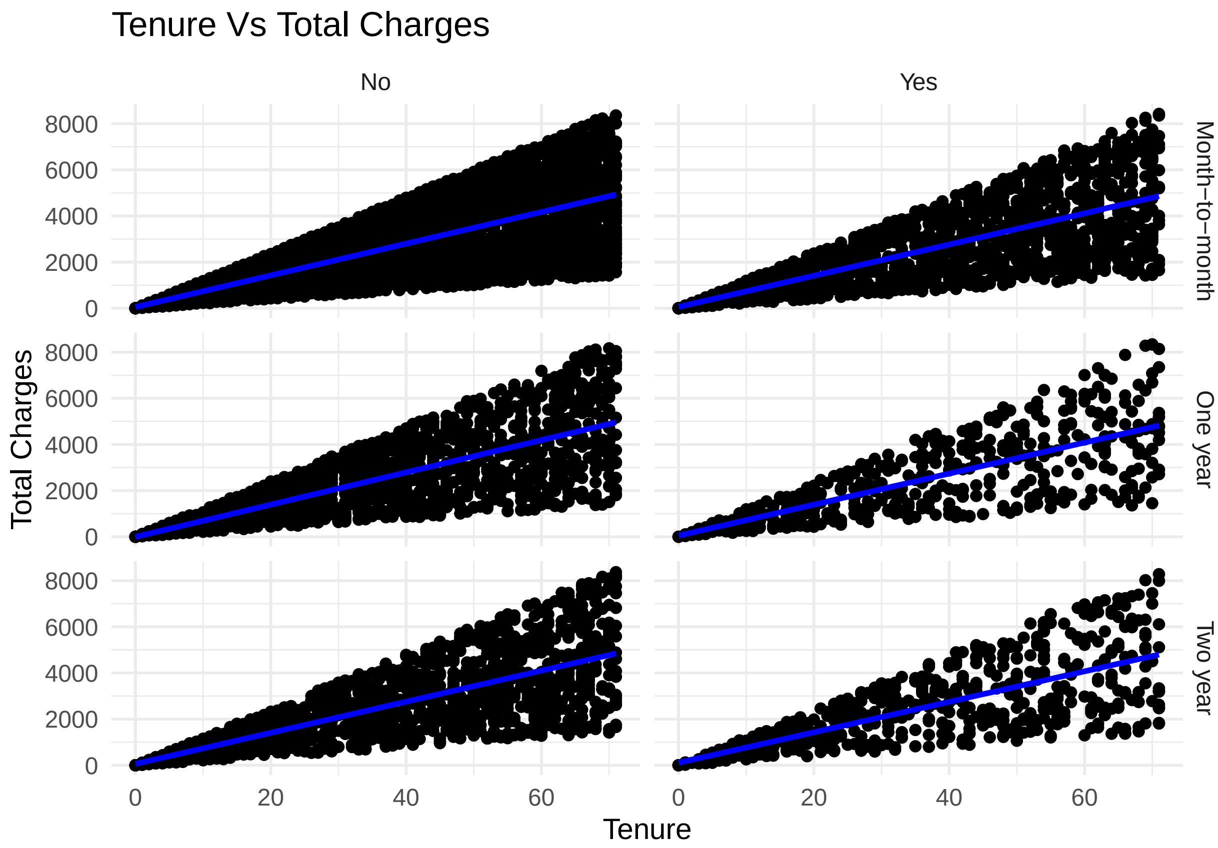
```
internetServ <- ggplot(customer_churn,aes(x=InternetService))+
  geom_bar()+
  labs(
    title = "Freq. per Internet Service"
  )+
  xlab("Internet Service")+
  ylab("Frquency")+
  theme_minimal()

contra <- ggplot(customer_churn,aes(x=Contract))+
  geom_bar()+
  labs(
    title = "Freq. per Contract"
  )+
  xlab("Contract")+
  ylab("Frquency")+
  theme_minimal()
barplts <- grid.arrange(internetServ,contra,ncol=2)
```




```
pltTenureVTotalCharge <- ggplot(customer_churn,aes(x=Tenure,y=TotalCharges))+
  geom_point()+
  geom_smooth(method = "lm",formula = y ~ x, color = "blue", se = FALSE,fullrange = TRUE) +
  labs(
    title = "Tenure Vs Total Charges"
  )+
```

```
  xlab("Tenure")+
  ylab("Total Charges")+
  facet_grid(Contract ~ Churn) +
  theme_minimal()
pltTenureVTotalCharge
```




```
# Removing Outliers

Q1 <- quantile(customer_churn$TotalCharges, 0.25, na.rm = TRUE)
Q3 <- quantile(customer_churn$TotalCharges, 0.75, na.rm = TRUE)

IQR_value <- Q3 - Q1

lower_bound <- Q1 - 1.5 * IQR_value
upper_bound <- Q3 + 1.5 * IQR_value

customer_churn_removedOutliers <- customer_churn[customer_churn$TotalCharges >= lower_bound & customer_c

#Standardizing and Normalizing Numerical Data

customer_churn_removedOutliers <- customer_churn_removedOutliers %>%
  mutate(
    zScoreTenure = (Tenure - mean(Tenure))/sd(Tenure),
    zScoreMonthlyCharges = (MonthlyCharges - mean(MonthlyCharges))/sd(MonthlyCharges),
    normalTotalCharges = (TotalCharges - min(TotalCharges)) / (max(TotalCharges) - min(TotalCharges))
```

```
  )
customer_churn_removedOutliers$SeniorCitizen <- ifelse(customer_churn_removedOutliers$SeniorCitizen == 
customer_churn_removedOutliers <- customer_churn_removedOutliers[, !names(customer_churn_removedOutliers

# Turning Categorical Data to Factors

customer_churn_removedOutliers$Gender <- as.factor(customer_churn_removedOutliers$Gender)
customer_churn_removedOutliers$SeniorCitizen <- as.factor(customer_churn_removedOutliers$SeniorCitizen)
customer_churn_removedOutliers$Partner <- as.factor(customer_churn_removedOutliers$Partner)
customer_churn_removedOutliers$Dependents <- as.factor(customer_churn_removedOutliers$Dependents)
customer_churn_removedOutliers$PhoneService <- as.factor(customer_churn_removedOutliers$PhoneService)
customer_churn_removedOutliers$InternetService <- as.factor(customer_churn_removedOutliers$InternetServi
customer_churn_removedOutliers$Contract <- as.factor(customer_churn_removedOutliers$Contract)
customer_churn_removedOutliers$Churn <- as.factor(customer_churn_removedOutliers$Churn)
```

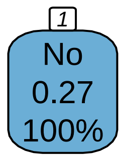
Remark:

- The distribution for Tenure and Monthly Charges have a relatively normal distribution. The distribution of the Total Charges is right-skewed with extreme outliers, which would require normalization.
- The most popular contract type is month-to-month as this offers flexibility amongst the customers, this is especially evident given the data points of the churned customers (3rd plot, right-side).
- A positive relationship between total charges and tenure is seen as the more a customer stays within the contract the more they have to pay for their monthly bills.
- To get rid of extreme outliers capping it into the IQR was performed.
- Given that Tenure and Monthly Charges are relatively normally distributed, their z-scores were obtained, while total charges was normalized using min-max normalization as the data was skewed.

```
# Spliting the Data
train_index_tree <- createDataPartition(customer_churn_removedOutliers$Churn, p = 0.8, list = FALSE)
train_data_tree <- customer_churn_removedOutliers[train_index_tree, ]
test_data_tree  <- customer_churn_removedOutliers[-train_index_tree, ]

customer_churnLogisModel <- customer_churn_removedOutliers %>%
  mutate(Churn = ifelse(Churn == "Yes", 1, 0))
train_index_logis <- createDataPartition(customer_churn_removedOutliers$Churn, p = 0.8, list = FALSE)
train_data_logis <- customer_churnLogisModel[train_index_logis, ]
test_data_logis  <- customer_churnLogisModel[-train_index_logis, ]
```

```
tree_model <- rpart(Churn ~ .,
                    data = train_data_tree,
                    method = "class")
rpart.plot(tree_model, box.palette = "auto", nn = TRUE)
```




```r
summary(tree_model)
```

```
## Call:
## rpart(formula = Churn ~ ., data = train_data_tree, method = "class")
##   n= 7943
##
##   CP nsplit rel error xerror xstd
## 1  0      0         1      0    0
##
## Node number 1: 7943 observations
##   predicted class=No  expected loss=0.2710563  P(node) =1
##     class counts:  5790  2153
##    probabilities: 0.729 0.271
```

```r
logistic_model <- glm(Churn ~ .,
                      data = train_data_logis,
                      family = "binomial")
summary(logistic_model)
```

```
##
## Call:
## glm(formula = Churn ~ ., family = "binomial", data = train_data_logis)
##
## Coefficients:
```

```
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)                -0.76119    0.14243  -5.344 9.08e-08 ***
## GenderMale                 -0.08156    0.05062  -1.611   0.1072
## SeniorCitizenYes           -0.01398    0.07044  -0.198   0.8427
## PartnerYes                 -0.06572    0.05057  -1.300   0.1937
## DependentsYes              -0.07796    0.05568  -1.400   0.1615
## PhoneServiceYes            -0.05413    0.08463  -0.640   0.5224
## InternetServiceFiber optic -0.01559    0.05631  -0.277   0.7819
## InternetServiceNo          -0.11066    0.07009  -1.579   0.1144
## ContractOne year           -0.11336    0.06664  -1.701   0.0889 .
## ContractTwo year            0.07076    0.06457   1.096   0.2731
## zScoreTenure                0.03438    0.06589   0.522   0.6019
## zScoreMonthlyCharges        0.01212    0.05022   0.241   0.8093
## normalTotalCharges         -0.14494    0.32434  -0.447   0.6550
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 9282.3  on 7942  degrees of freedom
## Residual deviance: 9267.5  on 7930  degrees of freedom
## AIC: 9293.5
##
## Number of Fisher Scoring iterations: 4
```

Decision Trees:

- Decision Trees optimizes the Gini Impunity of given features, it is a way in asignning a probability in misclassifying at a given node.
- Decision Trees are Non-linear meaning there is no assumption of linearity and non-parametric meaning it would be applicable to a wide range of problems in machine learning.
- As the data set gets bigger the more the Decision Tree model is prone to over fitting as it makes more nodes, to which optimizing the tree by cutting-off given nodes.

Logistic Regression:

- The logistic regression uses the sigmoid function in order to create the model. Logistic regression aims to find the maximum likelihood function or the parameters beta, which is done through gradient descent or Newton's Method.
- Logistic regression is a non-parametric approach as it has no assumption on the given distribution, however what logistic regression needs is an assumption of linearity, meaning we can represent the given data into a linear combination.

Bias-Variance Trade-Off

- In the models being trained, bias refers to the performance of the model when predicting test data, having a high bias shows that the model is very flexible and is not able to capture the true behavior of the given data, this is also known as underfitting in model building. Variance is the concept of overfitting, where the model performs very well on the training data as it prioritizes their actual values and fails to generalize it's behavior. *Having this balance between bias and variance is important as we would like a model that is able to capture the general behavior of the given data but at the same time generalized enough so that when new data points will be introduced it can give a good prediction.

```r
train_control <- trainControl(method="cv", number=10)
cv_results_dt <- train(Churn ~ ., data=train_data_tree, method="rpart", trControl=train_control)
cv_results_lg <- train(Churn~.,data = train_data_tree, method="glm", family="binomial", trControl=train_

print(cv_results_dt)
```

```
## CART
##
## 7943 samples
##   10 predictor
##    2 classes: 'No', 'Yes'
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 7149, 7148, 7149, 7148, 7148, 7149, ...
## Resampling results across tuning parameters:
##
##   cp            Accuracy   Kappa
##   0.0003715745  0.6741833  -0.01047852
##   0.0004644682  0.6834970  -0.01096740
##   0.0005109150  0.6834970  -0.01204600
##
## Accuracy was used to select the optimal model using the largest value.
## The final value used for the model was cp = 0.000510915.
```

```r
print(cv_results_lg)
```

```
## Generalized Linear Model
##
## 7943 samples
##   10 predictor
##    2 classes: 'No', 'Yes'
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 7149, 7149, 7149, 7149, 7148, 7149, ...
## Resampling results:
##
##   Accuracy  Kappa
##   0.728944  0
```

Remark: The decision tree was tuned by the different complexity parameter (cp) where cp=0.0006192909. With this the created decision tree model has an accuracy of 71.5% with a Kappa close to 0. For the Logistic Regression the given accuracy is 72.89%, this shows that even without tuning the logistic regression performed better than the decision tree model. However, it is still important to note that the model is 0, this suggests that the model fails to provide meaningful agreement beyond random chance.

```r
dt_preds <- predict(cv_results_dt, test_data_tree)
lg_preds <- predict(cv_results_lg, test_data_tree)

dt_cm <- confusionMatrix(dt_preds, test_data_tree$Churn)
print(dt_cm)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction   No  Yes
##        No  1447  538
##        Yes    0    0
##
##                Accuracy : 0.729
##                  95% CI : (0.7088, 0.7484)
##     No Information Rate : 0.729
##     P-Value [Acc > NIR] : 0.5116
##
##                   Kappa : 0
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 1.000
##             Specificity : 0.000
##          Pos Pred Value : 0.729
##          Neg Pred Value :   NaN
##              Prevalence : 0.729
##          Detection Rate : 0.729
##    Detection Prevalence : 1.000
##       Balanced Accuracy : 0.500
##
##        'Positive' Class : No
##
```

```
lg_cm <- confusionMatrix(lg_preds, test_data_tree$Churn)
print(lg_cm)
```


```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction   No  Yes
##        No  1447  538
##        Yes    0    0
##
##                Accuracy : 0.729
##                  95% CI : (0.7088, 0.7484)
##     No Information Rate : 0.729
##     P-Value [Acc > NIR] : 0.5116
##
##                   Kappa : 0
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 1.000
##             Specificity : 0.000
##          Pos Pred Value : 0.729
##          Neg Pred Value :   NaN
##              Prevalence : 0.729
##          Detection Rate : 0.729
##    Detection Prevalence : 1.000
```

```
##        Balanced Accuracy : 0.500
##
##         'Positive' Class : No
##
```

```
dt_accuracy <- dt_cm$overall["Accuracy"]
dt_precision <- dt_cm$byClass["Pos Pred Value"]  # Precision
dt_recall <- dt_cm$byClass["Sensitivity"]  # Recall
dt_f1 <- 2 * (dt_precision * dt_recall) / (dt_precision + dt_recall)

lg_accuracy <- lg_cm$overall["Accuracy"]
lg_precision <- lg_cm$byClass["Pos Pred Value"]
lg_recall <- lg_cm$byClass["Sensitivity"]
lg_f1 <- 2 * (lg_precision * lg_recall) / (lg_precision + lg_recall)

cat("DT Metrics\nAcuracy: ",dt_accuracy,"\nPercision: ",dt_precision,"\nRecall: ", dt_recall,"\nF1 Score
```

```
## DT Metrics
## Acuracy:  0.7289673
## Percision:  0.7289673
## Recall:  1
## F1 Score:  0.8432401
```

```
cat("\n\nLG Metrics\nAcuracy: ",lg_accuracy,"\nPercision: ",lg_precision,"\nRecall: ", lg_recall,"\nF1 S
```

```
##
##
## LG Metrics
## Acuracy:  0.7289673
## Percision:  0.7289673
## Recall:  1
## F1 Score:  0.8432401
```

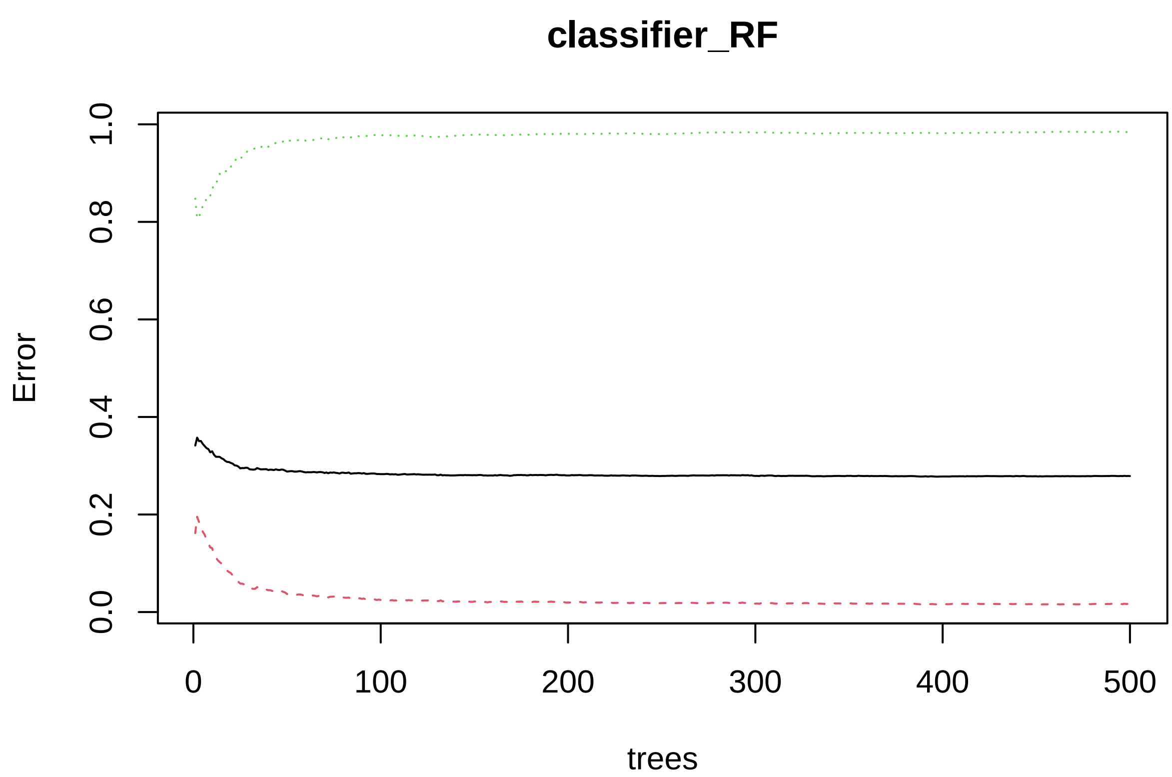
Remarks: The metrics for both of the decision tree and logistic regression is the same. Accuracy of both models are high, implying that most of the labels are predicted correctly by the model. Given a high value of percision implying that there are less false positives.Given a recall score of 1 shows that the model correctly identifies costumers that can churn. Lastly the high F1 score shows that both models are well balanced where it does not produce much false positives and false negatives.

```
classifier_RF <- randomForest(Churn ~ ., data=train_data_tree,proximity=TRUE)
summary(classifier_RF)
```

```
##                 Length   Class  Mode
## call                4 -none- call
## type                1 -none- character
## predicted        7943 factor numeric
## err.rate         1500 -none- numeric
## confusion           6 -none- numeric
## votes           15886 matrix numeric
## oob.times        7943 -none- numeric
## classes             2 -none- character
## importance         10 -none- numeric
```

```
## importanceSD            0 -none- NULL
## localImportance         0 -none- NULL
## proximity        63091249 -none- numeric
## ntree                   1 -none- numeric
## mtry                    1 -none- numeric
## forest                 14 -none- list
## y                    7943 factor numeric
## test                    0 -none- NULL
## inbag                   0 -none- NULL
## terms                   3 terms  call
```

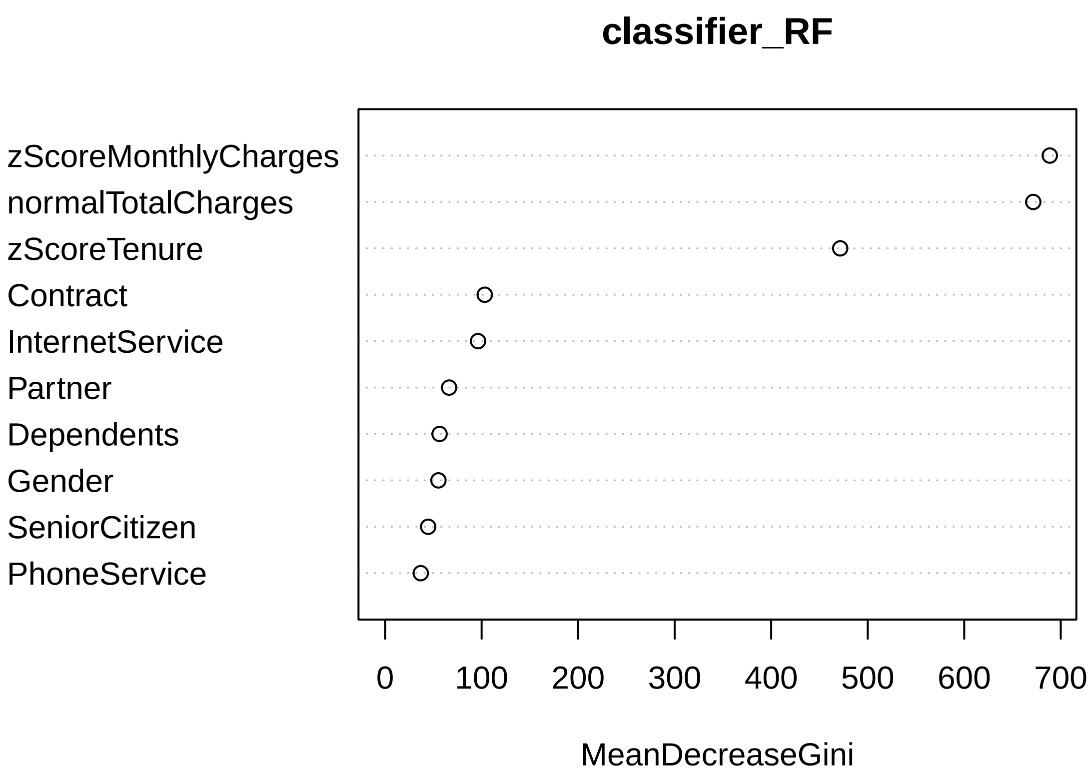
```r
plot(classifier_RF)
```




```r
importance(classifier_RF)
```

```
##                         MeanDecreaseGini
## Gender                          55.24370
## SeniorCitizen                   44.63751
## Partner                         66.30921
## Dependents                      56.43376
## PhoneService                    36.91690
## InternetService                 96.28896
## Contract                       103.25502
## zScoreTenure                   471.49287
## zScoreMonthlyCharges           688.63164
## normalTotalCharges             671.49953
```

```r
varImpPlot(classifier_RF)
```




```r
grid <- expand.grid(mtry = c(2, 3, 4, 5, 6, 7, 8, 9, 10))

rfGridSearch <- train(Churn ~ .,
                      data = train_data_tree,
                      method = "rf",
                      trControl = train_control,
                      tuneGrid = grid)
print(rfGridSearch)
```

```
## Random Forest
##
## 7943 samples
##   10 predictor
##    2 classes: 'No', 'Yes'
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 7148, 7149, 7149, 7149, 7148, 7149, ...
## Resampling results across tuning parameters:
##
##   mtry  Accuracy   Kappa
##    2    0.7289440   0.000000000
##    3    0.7269292  -0.001042160
```

```
##    4    0.7193765   0.007829700
##    5    0.7121995   0.015891109
##    6    0.7051501   0.004710267
##    7    0.7050219   0.005677530
##    8    0.7033862   0.003100852
##    9    0.7046446   0.009986925
##   10    0.7026307   0.008976551
##
## Accuracy was used to select the optimal model using the largest value.
## The final value used for the model was mtry = 2.
```

```
predictionGrid <- predict(rfGridSearch, newdata = test_data_tree)
rf_cm<-confusionMatrix(predictionGrid, test_data_tree$Churn)

rf_accuracy <- rf_cm$overall["Accuracy"]
rf_precision <- rf_cm$byClass["Pos Pred Value"]
rf_recall <- rf_cm$byClass["Sensitivity"]
rf_f1 <- 2 * (rf_precision * rf_recall) / (rf_precision + rf_recall)

cat("RF Metrics\nAcuracy: ",rf_accuracy,"\nPercision: ",rf_precision,"\nRecall: ", rf_recall,"\nF1 Scor
```

```
## RF Metrics
## Acuracy:  0.7289673
## Percision:  0.7289673
## Recall:  1
## F1 Score:  0.8432401
```

Remarks:

- The Plot of model shows that the model stabalizes at 100-200 trees implying adding more trees would not contribute much to the over all performance of the mode.
- The mean decrease gini shows that importance of each data variable, which shows the most important predictors are zScoreMonthlyCharges, normalTotalCharges, zScoreTenure, InternetService, Contract.
- The grid search was able to identify the best mtry or number of features randomly chosen which is equivalent to 2.
- In terms of Kappa the largest mtry is 9 however the best tune is where kappa is 2.739130e-03 while mtry=3 with accuracy that is near mtry = 2.
- Accuracy and Percision of the model is the same. High accuracy means that the model correctly labels the data. The Percision is high indicating that the model is likely to produce false positives.
- The high F1 Score shows that the model is over all balanced, indicating that the model does not produce much false positives and false negatives.
- The low kappa, the high error on the churn of Yes shows that thier is an imbalance within the data where there are more data points for customers that have not churned stead of those that was.

```
customer_churnMain <- customer_churn[, !names(customer_churn) %in% c("CustomerID")] %>%
  mutate(Churn = ifelse(Churn == "Yes", 1, 0))
trainMainInd <- createDataPartition(customer_churnMain$Churn, p = 0.8, list = FALSE)
trainMain <- customer_churnMain[trainMainInd, ]
testMain  <- customer_churnMain[-trainMainInd, ]

logistic_model <- glm(Churn ~ Tenure+MonthlyCharges+TotalCharges,
                      data = trainMain,
                      family = "binomial")
summary(logistic_model)
```

```
##
## Call:
## glm(formula = Churn ~ Tenure + MonthlyCharges + TotalCharges,
##     family = "binomial", data = trainMain)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.052e+00  1.320e-01  -7.964 1.67e-15 ***
## Tenure          3.235e-03  3.153e-03   1.026    0.305
## MonthlyCharges  8.908e-05  1.732e-03   0.051    0.959
## TotalCharges   -2.791e-05  4.194e-05  -0.665    0.506
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 9316.2  on 7999  degrees of freedom
## Residual deviance: 9313.4  on 7996  degrees of freedom
## AIC: 9321.4
##
## Number of Fisher Scoring iterations: 4
```

Remarks:

- As tenure increase the likelihood of tenure increases although slightly, it also shows that it is not-significant as it is greater than 0.05.
- As Monthly Charges increase the likelihood of tenure increases although slightly, it also shows that it is not-significant as it is greater than 0.05.
- As Monthly Charges increase the likelihood of tenure decreases although slightly, it also shows that it is not-significant as it is greater than 0.05.
- Given that the Residual Deviance is high it implys that the model does not explain the data well, in addition the high AIC shows that the model might be either overfitting or underfitting, in otherwords not balanced.

High Dimentionality:

- High dimensionality is where there are so many features within the data that the data gets split too far apart, this is especially a problem for linear and distance based models like KNN as the points are too far to infer a behavior for the data, in addition it creates computational problems as the more dimensions the more time the models need to calculate the given coefficients.

```r
churnComponents <- customer_churnMain[, c("Tenure", "MonthlyCharges", "TotalCharges")]
pcaResults <- prcomp(churnComponents, scale = TRUE)
pcaResults$rotation <- -1*pcaResults$rotation
pcaResults$rotation
```

```
##                      PC1          PC2        PC3
## Tenure         0.5794279 -0.573333342  0.5792687
## MonthlyCharges 0.3970680  0.819264829  0.4136933
## TotalCharges   0.7117586  0.009696324 -0.7023572
```

```r
summary(pcaResults)
```

```
## Importance of components:
##                           PC1    PC2     PC3
## Standard deviation     1.3862 1.0123 0.23130
## Proportion of Variance 0.6405 0.3416 0.01783
## Cumulative Proportion  0.6405 0.9822 1.00000
```

Remarks:

- The PCA shows 3 components, for PC1 it reveals the general trend for all three variables, for PC2 shows a negative relationship between tenure and monthly charges, similarly PC3 shows the negative relationship between total charges and tenure.
- The summary shows that by the cumulative proportion PC1 and PC2 is enough to explain 99.22% of the data, given the results of this PCA it shows that we have reduced the dimension.

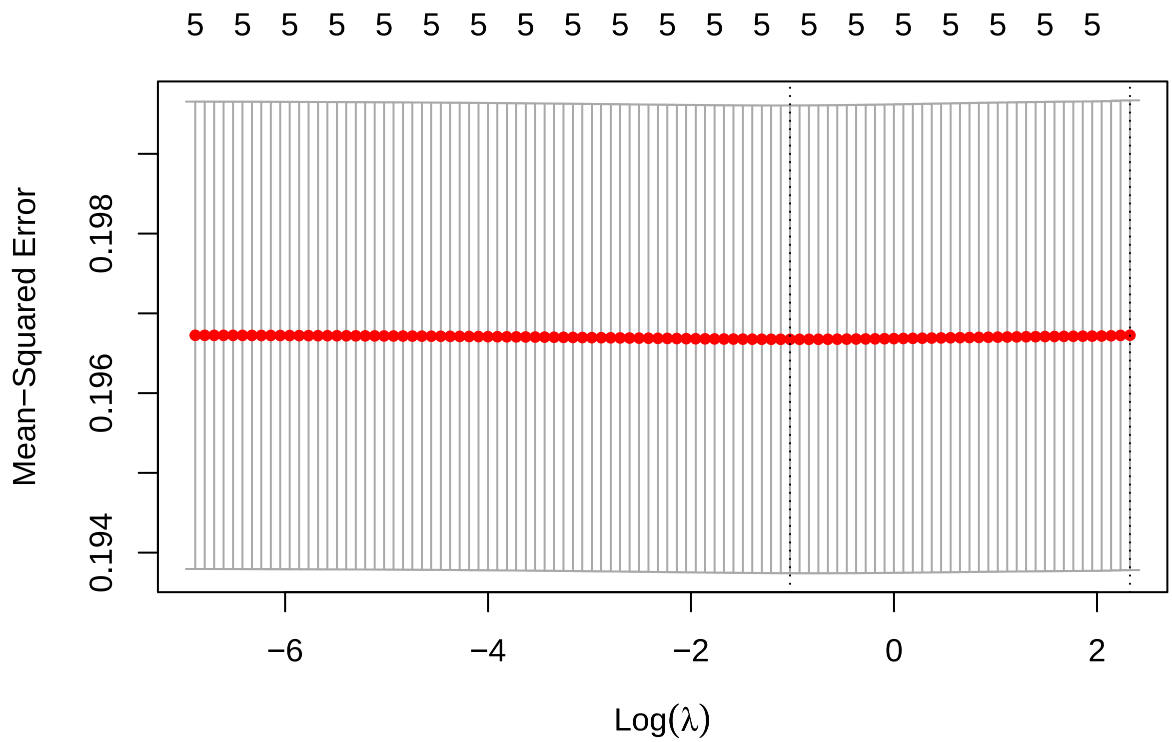
```r
X <- model.matrix(Churn ~ Tenure + MonthlyCharges + TotalCharges+Contract, data = trainMain)[, -1]
y <- as.numeric(trainMain$Churn)
ridgemodel <- glmnet(X, y, alpha = 0)
summary(ridgemodel)
```

```
##           Length Class     Mode
## a0        100    -none-    numeric
## beta      500    dgCMatrix S4
## df        100    -none-    numeric
## dim         2    -none-    numeric
## lambda    100    -none-    numeric
## dev.ratio 100    -none-    numeric
## nulldev     1    -none-    numeric
## npasses     1    -none-    numeric
## jerr        1    -none-    numeric
## offset      1    -none-    logical
## call        4    -none-    call
## nobs        1    -none-    numeric
```

```r
cv_modelRidge <- cv.glmnet(X, y, alpha = 0)
best_lambda <- cv_modelRidge$lambda.min
best_lambda
```

```
## [1] 0.3590108
```

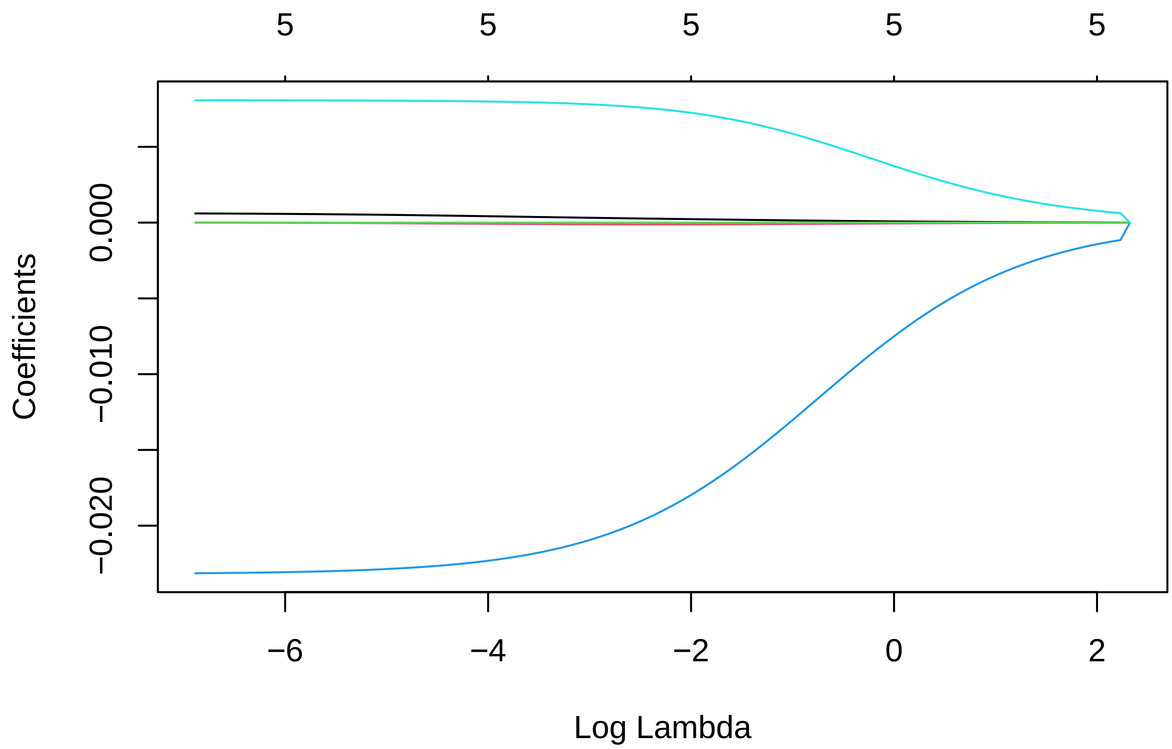
```r
plot(cv_modelRidge)
```




```
best_model_ridge <- glmnet(X, y, alpha = 0, lambda = best_lambda)
coef(best_model_ridge)
```

```
## 6 x 1 sparse Matrix of class "dgCMatrix"
##                              s0
## (Intercept)        2.725361e-01
## Tenure             1.491293e-04
## MonthlyCharges    -9.608203e-05
## TotalCharges      -2.134111e-07
## ContractOne year -1.316763e-02
## ContractTwo year  5.913566e-03
```

```
plot(ridgemodel, xvar = "lambda")
```



Remark:

- Optimal Lambda identified throigh Cross Validation is 0.8663789

```
modelLasso <- glmnet(X, y, alpha = 0)
summary(modelLasso)
```

```
##           Length Class     Mode
## a0        100    -none-    numeric
## beta      500    dgCMatrix S4
## df        100    -none-    numeric
## dim         2    -none-    numeric
## lambda    100    -none-    numeric
## dev.ratio 100    -none-    numeric
## nulldev     1    -none-    numeric
## npasses     1    -none-    numeric
## jerr        1    -none-    numeric
## offset      1    -none-    logical
## call        4    -none-    call
## nobs        1    -none-    numeric
```

```
coef(modelLasso)
```

```
## 6 x 100 sparse Matrix of class "dgCMatrix"
```

```
##   [[ suppressing 100 column names 's0', 's1', 's2' ... ]]
```

```
##
## (Intercept)       2.690000e-01  2.692684e-01  2.692938e-01  2.693216e-01
## Tenure            2.661216e-40  1.192618e-05  1.302729e-05  1.422402e-05
## MonthlyCharges   -1.865285e-40 -8.382806e-06 -9.158548e-06 -1.000183e-05
## TotalCharges      1.241919e-43  3.992201e-09  4.202005e-09  4.399570e-09
## ContractOne year -2.565327e-38 -1.146406e-03 -1.251889e-03 -1.366451e-03
## ContractTwo year  1.424013e-38  6.271054e-04  6.838621e-04  7.453212e-04
##
## (Intercept)       2.693519e-01  2.693849e-01  2.694209e-01  2.694601e-01
## Tenure            1.552352e-05  1.693326e-05  1.846107e-05  2.011500e-05
## MonthlyCharges   -1.091771e-05 -1.191148e-05 -1.298862e-05 -1.415476e-05
## TotalCharges      4.578104e-09  4.729357e-09  4.843354e-09  4.908133e-09
## ContractOne year -1.490755e-03 -1.625492e-03 -1.771374e-03 -1.929132e-03
## ContractTwo year  8.117905e-04  8.835829e-04  9.610126e-04  1.044392e-03
##
## (Intercept)       2.695026e-01  2.695488e-01  2.695990e-01  2.696533e-01
## Tenure            2.190335e-05  2.383462e-05  2.591740e-05  2.816031e-05
## MonthlyCharges   -1.541568e-05 -1.677721e-05 -1.824521e-05 -1.982546e-05
## TotalCharges      4.909455e-09  4.830497e-09  4.651521e-09  4.349522e-09
## ContractOne year -2.099511e-03 -2.283263e-03 -2.481136e-03 -2.693870e-03
## ContractTwo year  1.134027e-03  1.230211e-03  1.333219e-03  1.443303e-03
##
## (Intercept)       2.697120e-01  2.697755e-01  2.698439e-01  2.699175e-01
## Tenure            3.057193e-05  3.316069e-05  3.593472e-05  3.890179e-05
## MonthlyCharges   -2.152359e-05 -2.334495e-05 -2.529454e-05 -2.737682e-05
## TotalCharges      3.897867e-09  3.265921e-09  2.418675e-09  1.316382e-09
## ContractOne year -2.922183e-03 -3.166757e-03 -3.428228e-03 -3.707166e-03
## ContractTwo year  1.560681e-03  1.685532e-03  1.817985e-03  1.958111e-03
##
## (Intercept)       2.699965e-01  2.700811e-01  2.701715e-01  2.702677e-01
## Tenure            4.206912e-05  4.544324e-05  4.902989e-05  5.283638e-05
## MonthlyCharges   -2.959559e-05 -3.195381e-05 -3.445347e-05 -3.709434e-05
## TotalCharges     -8.580301e-11 -1.838142e-09 -3.996611e-09 -6.626164e-09
## ContractOne year -4.004066e-03 -4.319323e-03 -4.653218e-03 -5.005880e-03
## ContractTwo year  2.105914e-03  2.261322e-03  2.424178e-03  2.594236e-03
##
## (Intercept)       2.703699e-01  2.704780e-01  2.705921e-01  2.707119e-01
## Tenure            5.686205e-05  6.111132e-05  6.558533e-05  7.028378e-05
## MonthlyCharges   -3.987755e-05 -4.280022e-05 -4.585850e-05 -4.904671e-05
## TotalCharges     -9.790885e-09 -1.356724e-08 -1.803650e-08 -2.328648e-08
## ContractOne year -5.377347e-03 -5.767439e-03 -6.175811e-03 -6.601928e-03
## ContractTwo year  2.771146e-03  2.954462e-03  3.143631e-03  3.337999e-03
##
## (Intercept)       2.708372e-01  2.709677e-01  2.711028e-01  2.712420e-01
## Tenure            7.520490e-05  8.034540e-05  8.570052e-05  9.126406e-05
## MonthlyCharges   -5.235714e-05 -5.578003e-05 -5.930345e-05 -6.291331e-05
## TotalCharges     -2.941161e-08 -3.651281e-08 -4.469742e-08 -5.407904e-08
## ContractOne year -7.045058e-03 -7.504260e-03 -7.978391e-03 -8.466110e-03
## ContractTwo year  3.536810e-03  3.739217e-03  3.944293e-03  4.151041e-03
##
## (Intercept)       2.713847e-01  2.715300e-01  2.716771e-01  2.718249e-01
## Tenure            9.702851e-05  1.029852e-04  1.091245e-04  1.154360e-04
## MonthlyCharges   -6.659343e-05 -7.032559e-05 -7.408969e-05 -7.786401e-05
## TotalCharges     -6.477734e-08 -7.691785e-08 -9.063163e-08 -1.060551e-07
```

```
## ContractOne year -8.965882e-03 -9.476000e-03 -9.994602e-03 -1.051970e-02
## ContractTwo year  4.358413e-03  4.565329e-03  4.770699e-03  4.973444e-03
##
## (Intercept)       2.719722e-01  2.721179e-01  2.722608e-01  2.723994e-01
## Tenure            1.219089e-04  1.285324e-04  1.352955e-04  1.421882e-04
## MonthlyCharges   -8.162539e-05 -8.534958e-05 -8.901156e-05 -9.258592e-05
## TotalCharges     -1.233294e-07 -1.426006e-07 -1.640183e-07 -1.877362e-07
## ContractOne year -1.104919e-02 -1.158092e-02 -1.211269e-02 -1.264231e-02
## ContractTwo year  5.172519e-03  5.366935e-03  5.555777e-03  5.738224e-03
##
## (Intercept)       2.725324e-01  2.726584e-01  2.727760e-01  2.728839e-01
## Tenure            1.492007e-04  1.563249e-04  1.635535e-04  1.708810e-04
## MonthlyCharges   -9.604720e-05 -9.937027e-05 -1.025307e-04 -1.055051e-04
## TotalCharges     -2.139107e-07 -2.427003e-07 -2.742652e-07 -3.087660e-07
## ContractOne year -1.316762e-02 -1.368652e-02 -1.419704e-02 -1.469731e-02
## ContractTwo year  5.913563e-03  6.081197e-03  6.240654e-03  6.391588e-03
##
## (Intercept)       2.729807e-01  2.730652e-01  2.731362e-01  2.731925e-01
## Tenure            1.783035e-04  1.858188e-04  1.934265e-04  2.011277e-04
## MonthlyCharges   -1.082714e-04 -1.108090e-04 -1.130994e-04 -1.151258e-04
## TotalCharges     -3.463623e-07 -3.872120e-07 -4.314691e-07 -4.792825e-07
## ContractOne year -1.518562e-02 -1.566046e-02 -1.612047e-02 -1.656452e-02
## ContractTwo year  6.533777e-03  6.667117e-03  6.791618e-03  6.907386e-03
##
## (Intercept)       2.732331e-01  2.732573e-01  2.732641e-01  2.732529e-01
## Tenure            2.089253e-04  2.168234e-04  2.248270e-04  2.329422e-04
## MonthlyCharges   -1.168736e-04 -1.183305e-04 -1.194865e-04 -1.203337e-04
## TotalCharges     -5.307935e-07 -5.861336e-07 -6.454224e-07 -7.087643e-07
## ContractOne year -1.699166e-02 -1.740114e-02 -1.779243e-02 -1.816517e-02
## ContractTwo year  7.014617e-03  7.113580e-03  7.204604e-03  7.288065e-03
##
## (Intercept)       2.732233e-01  2.731749e-01  2.731663e-01  2.731063e-01
## Tenure            2.411752e-04  2.495326e-04  2.567029e-04  2.647126e-04
## MonthlyCharges   -1.208667e-04 -1.210823e-04 -1.216371e-04 -1.215255e-04
## TotalCharges     -7.762460e-07 -8.479337e-07 -9.101250e-07 -9.835713e-07
## ContractOne year -1.851918e-02 -1.885445e-02 -1.917187e-02 -1.947026e-02
## ContractTwo year  7.364374e-03  7.433960e-03  7.497242e-03  7.554760e-03
##
## (Intercept)       2.730170e-01  2.729057e-01  2.727751e-01  0.2726263323
## Tenure            2.730820e-04  2.816598e-04  2.903946e-04  0.0002992669
## MonthlyCharges   -1.209905e-04 -1.201146e-04 -1.189312e-04 -0.0001174590
## TotalCharges     -1.063455e-06 -1.148165e-06 -1.237083e-06 -0.0000013299
## ContractOne year -1.975065e-02 -2.001354e-02 -2.025950e-02 -0.0204891482
## ContractTwo year  7.606870e-03  7.654008e-03  7.696589e-03  0.0077350030
##
## (Intercept)       0.2724604230  2.722781e-01  2.720804e-01  2.718680e-01
## Tenure            0.0003082667  3.173863e-04  3.266175e-04  3.359502e-04
## MonthlyCharges   -0.0001157124 -1.137052e-04 -1.114516e-04 -1.089670e-04
## TotalCharges     -0.0000014264 -1.526377e-06 -1.629614e-06 -1.735868e-06
## ContractOne year -0.0207031653 -2.090226e-02 -2.108715e-02 -2.125860e-02
## ContractTwo year  0.0077696207  7.800787e-03  7.828822e-03  7.854022e-03
##
## (Intercept)       2.716420e-01  2.714035e-01  0.2711536487  2.708937e-01
## Tenure            3.453727e-04  3.548707e-04  0.0003644284  3.740276e-04
```

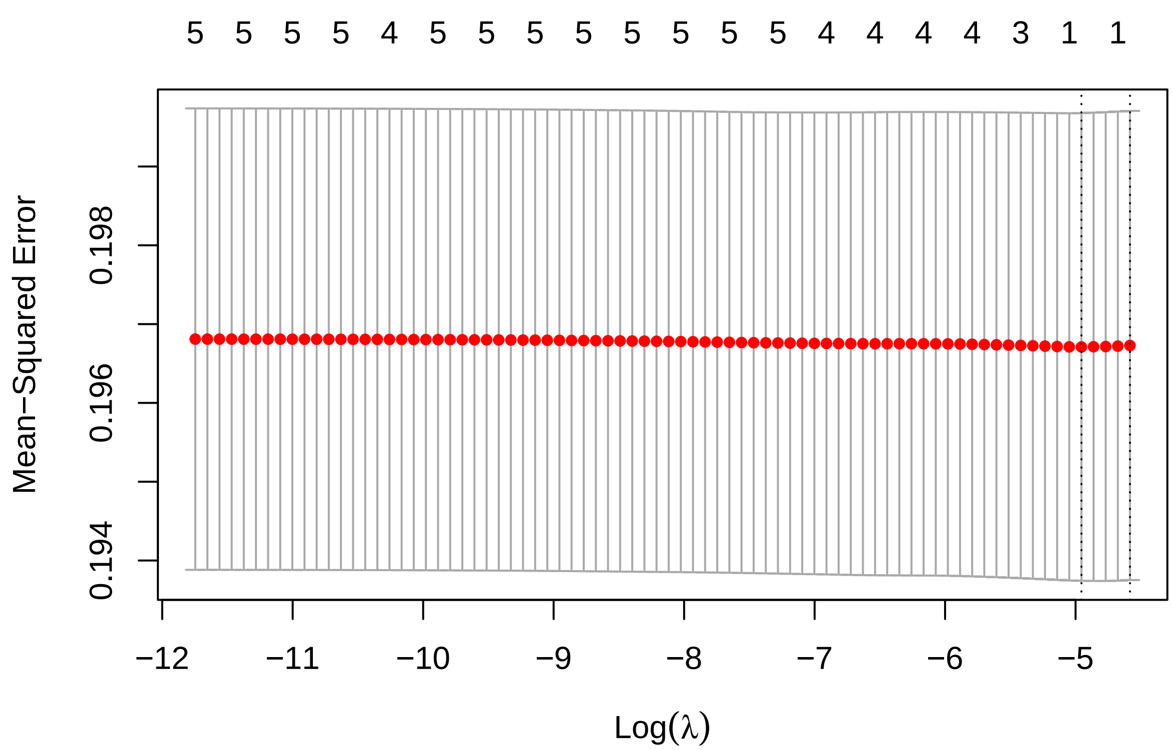
```
## MonthlyCharges   -1.062677e-04 -1.033710e-04 -0.0001002949 -9.705841e-05
## TotalCharges     -1.844868e-06 -1.956319e-06 -0.0000020699 -2.185268e-06
## ContractOne year -2.141734e-02 -2.156412e-02 -0.0216996592 -2.182468e-02
## ContractTwo year  7.876661e-03  7.896986e-03  0.0079152263  7.931588e-03
##
## (Intercept)       2.706248e-01  2.703484e-01  2.700658e-01  2.697783e-01
## Tenure            3.836488e-04  3.932708e-04  4.028717e-04  4.124284e-04
## MonthlyCharges   -9.368071e-05 -9.018152e-05 -8.658066e-05 -8.289796e-05
## TotalCharges     -2.302063e-06 -2.419910e-06 -2.538424e-06 -2.657216e-06
## ContractOne year -2.193987e-02 -2.204590e-02 -2.214340e-02 -2.223298e-02
## ContractTwo year  7.946260e-03  7.959412e-03  7.971197e-03  7.981754e-03
##
## (Intercept)       2.694872e-01  2.691939e-01  2.688996e-01  2.686056e-01
## Tenure            4.219175e-04  4.313155e-04  4.405992e-04  4.497458e-04
## MonthlyCharges   -7.915301e-05 -7.536506e-05 -7.155278e-05 -6.773414e-05
## TotalCharges     -2.775895e-06 -2.894074e-06 -3.011376e-06 -3.127434e-06
## ContractOne year -2.231521e-02 -2.239065e-02 -2.245981e-02 -2.252316e-02
## ContractTwo year  7.991210e-03  7.999675e-03  8.007252e-03  8.014033e-03
##
## (Intercept)       2.683132e-01  2.680233e-01  2.677372e-01  2.674557e-01
## Tenure            4.587334e-04  4.675413e-04  4.761500e-04  4.845418e-04
## MonthlyCharges   -6.392626e-05 -6.014529e-05 -5.640627e-05 -5.272310e-05
## TotalCharges     -3.241901e-06 -3.354447e-06 -3.464766e-06 -3.572578e-06
## ContractOne year -2.258117e-02 -2.263425e-02 -2.268280e-02 -2.272718e-02
## ContractTwo year  8.020099e-03  8.025525e-03  8.030377e-03  8.034714e-03
##
## (Intercept)       2.671797e-01  2.669101e-01  2.666476e-01  2.663928e-01
## Tenure            4.927006e-04  5.006122e-04  5.082643e-04  5.156468e-04
## MonthlyCharges   -4.910841e-05 -4.557357e-05 -4.212859e-05 -3.878220e-05
## TotalCharges     -3.677630e-06 -3.779698e-06 -3.878589e-06 -3.974139e-06
## ContractOne year -2.276775e-02 -2.280480e-02 -2.283863e-02 -2.286952e-02
## ContractTwo year  8.038591e-03  8.042055e-03  8.045150e-03  8.047914e-03
##
## (Intercept)       2.661462e-01  2.659082e-01  2.656792e-01  2.654594e-01
## Tenure            5.227514e-04  5.295721e-04  5.361045e-04  5.423465e-04
## MonthlyCharges   -3.554180e-05 -3.241351e-05 -2.940220e-05 -2.651157e-05
## TotalCharges     -4.066214e-06 -4.154711e-06 -4.239553e-06 -4.320694e-06
## ContractOne year -2.289771e-02 -2.292343e-02 -2.294689e-02 -2.296829e-02
## ContractTwo year  8.050384e-03  8.052589e-03  8.054558e-03  8.056316e-03
##
## (Intercept)       2.652491e-01  2.650483e-01  2.648570e-01  2.646753e-01
## Tenure            5.482975e-04  5.539587e-04  5.593327e-04  5.644237e-04
## MonthlyCharges   -2.374420e-05 -2.110162e-05 -1.858441e-05 -1.619228e-05
## TotalCharges     -4.398111e-06 -4.471806e-06 -4.541803e-06 -4.608147e-06
## ContractOne year -2.298779e-02 -2.300558e-02 -2.302179e-02 -2.303656e-02
## ContractTwo year  8.057886e-03  8.059287e-03  8.060539e-03  8.061657e-03
##
## (Intercept)       2.645030e-01  2.643400e-01  2.641860e-01  2.640410e-01
## Tenure            5.692369e-04  5.737789e-04  5.780570e-04  5.820794e-04
## MonthlyCharges   -1.392413e-05 -1.177820e-05 -9.752085e-06 -7.842884e-06
## TotalCharges     -4.670897e-06 -4.730133e-06 -4.785945e-06 -4.838435e-06
## ContractOne year -2.305003e-02 -2.306229e-02 -2.307347e-02 -2.308365e-02
## ContractTwo year  8.062655e-03  8.063547e-03  8.064343e-03  8.065055e-03
##
```

```
## (Intercept)       2.639046e-01  2.637765e-01  2.636565e-01  2.635442e-01
## Tenure            5.858549e-04  5.893929e-04  5.927032e-04  5.957957e-04
## MonthlyCharges   -6.047244e-06 -4.361454e-06 -2.781514e-06 -1.303201e-06
## TotalCharges     -4.887715e-06 -4.933903e-06 -4.977125e-06 -5.017509e-06
## ContractOne year -2.309293e-02 -2.310138e-02 -2.310907e-02 -2.311608e-02
## ContractTwo year  8.065692e-03  8.066261e-03  8.066770e-03  8.067226e-03
##
## (Intercept)       2.634392e-01  2.633413e-01  2.632501e-01  2.631653e-01
## Tenure            5.986806e-04  6.013682e-04  6.038687e-04  6.061922e-04
## MonthlyCharges    7.786283e-08  1.366162e-06  2.566232e-06  3.682618e-06
## TotalCharges     -5.055186e-06 -5.090289e-06 -5.122951e-06 -5.153303e-06
## ContractOne year -2.312247e-02 -2.312828e-02 -2.313358e-02 -2.313841e-02
## ContractTwo year  8.067634e-03  8.067999e-03  8.068326e-03  8.068620e-03
```

```r
cv_modelLasso <- cv.glmnet(X, y, alpha = 1)
best_lambdaLasso <- cv_modelLasso$lambda.min
best_lambdaLasso
```

```
## [1] 0.007047529
```

```r
plot(cv_modelLasso)
```




```r
best_model_lasso <- glmnet(X, y, alpha = 0, lambda = best_lambdaLasso)
coef(best_model_lasso)
```

```
## 6 x 1 sparse Matrix of class "dgCMatrix"
##                              s0
## (Intercept)       2.669938e-01
## Tenure            4.998383e-04
## MonthlyCharges   -4.650173e-05
## TotalCharges     -3.772925e-06
## ContractOne year -2.284776e-02
## ContractTwo year  8.046818e-03
```

Remarks:

- Feature selection helps us identify any features that are corrolated with each other and identify which features are able to explain majority of the variance within the model. This is to ensure that the number of features are optimized to avoid having high dimensions, for having more dimensions limits the model on it's performance.
- The coefficeints shows that only total charges and contract of one year is of negative, as these features increase the more it is less likely to churn. Inaddition the most important feature within the lasso regression is the intercept.